NIRAJ NAIDU

MECHANICAL DESIGN ENGINEER

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SCAN OR CLICK FOR ADDITIONLA INFORMATION AND PORTFOLIO

A passionate and detail-oriented Mechanical Design Engineer with practical experience in 3D modeling, product design, and manufacturing processes. Skilled in SolidWorks, CATIA, and ANSYS, with a strong foundation in hydraulic system design, reverse engineering, and precision machining. Experienced in leading Formula Student projects, designing EV components, and developing industrial equipment such as gantry cranes, mold presses, and tube bending machines. I combine creativity with technical expertise to deliver accurate, efficient, and innovative mechanical solutions—whether in professional engineering environments or independent freelance projects.

EDUCATION & PROFESSIONAL EXPERIENCE

April 2025 – April 2027

Master of Science in Digital Technology and Management

FOM Hochschule für Oekonomie & Management, Germany

(Grade-1.5/5)

M.Sc. student in Digital Technology and Management at FOM University, combining mechanical engineering expertise with digital manufacturing, sustainable production, and advanced product development. Experienced in CAD, additive manufacturing, and precision machining, with a focus on developing efficient and resource-conscious engineering solutions.

- Digital Manufacturing: CAD and digital technologies for modern production.
- Sustainable Engineering: Reducing material waste through CNC and additive manufacturing.
- Product Design: SolidWorks, CATIA, topology optimization, and DFM.
- Additive Manufacturing: 3D printing and material-efficient production.

June 2024 – December 2024

Design Engineer

KC Engineering and Automation

Worked as a Mechanical Design Engineer specializing in hydraulic machinery and industrial equipment, covering CAD design, reverse engineering, product development, and manufacturing support.

- Worked as a SolidWorks Designer and Drafter for hydraulic machinery.
- Designed hydraulic systems including cylinders, presses, and jacks.
- Completed the reverse engineering of the SLB 1100 Gantry Crane.
- Contributed to tube-bending machines, mold presses, and horizontal moving jacks.
- Created detailed 3D CAD models and engineering drawings focused on precision and manufacturability.
- Prepared manufacturing documentation and worked with fabrication teams to support production and assembly.

June 2020 – September 2024

Bachelor of Technology in Mechanical Engineering

Symbiosis Institute of Technology, Pune, India

Completed B.Tech in Mechanical Engineering with experience in precision mechanical design, Formula Student engineering, and energy-system prototyping. I worked on mold components, industrial fixtures, and high-precision assemblies with tolerances of up to 5 microns. As part of a Formula Student team, I designed automotive components using SolidWorks and CATIA, supported carbon-fibre fabrication and 3D printing, and contributed to an EV battery system that ranked 15th among 72 teams nationally. I also developed and tested a low-cost hydrogen generation system using water electrolysis, gaining practical experience in mechanical design, prototyping, and performance evaluation.

January 2024 – May 2024

Design Intern

ASB International Pvt. Ltd.

Mandatory Internship for B.Tech Degree

Worked on precision mechanical design and manufacturing, focusing on mold components, industrial fixtures, and high-accuracy assemblies.

- Precision Mold Design: Developed 3D models of mold components and fixtures.
- High-Precision Assembly: Achieved tolerances of up to 5 microns for stable, accurate machining.
- Manufacturing Standards: Gained practical experience in machining requirements, standards, and design practices.

May 2023 – July 2024

Seminar / Final Year Project

Design of a Hydrogen Generator

Final Year Project for completion of B.Tech Degree

Developed a low-cost hydrogen generation system using water electrolysis, focusing on efficient gas production, mechanical design, and experimental validation.

- Designed the generator and electrode assembly using SolidWorks.
- Compared KOH and $NaHCO_3$ electrolytes for hydrogen generation.
- Fabricated and tested the prototype for efficiency, safety, and performance.
- Gained practical experience in electrolysis, prototyping, and mechanical design.

May 2022 – May 2023

Design Head

Wrench Wielders Racing, Formula Student Team

Gained hands-on experience in automotive design, fabrication, and EV development as part of a Formula Student engineering team.

- Member of the Formula Student Design Team. Since June 2020 - May 2023
- Designed automotive components using SolidWorks and CATIA.
- Supported fabrication through carbon-fiber bodywork and 3D printing.
- Designed an EV battery system, ranking 15th among 72 teams nationally.
- Competed in major events including SUPRA 2022 and Formula Student EV 2021.

Skills

- **SOLIDWORKS - Expert**
- **CATIA - Expert**
- **AutoCAD-2D - Expert**
- **CURA - Expert**
- **RD WORKS - Expert**
- **ANSYS - Intermediate**
- **3D-PRINTING - Expert**
- **Drafting - Expert**

Languages

- **English - C1**
- **German - A1, Persuing A2**
- **Hindi - B2**
- **Marathi - B2**

Certificates for SolidWorks, CATIA, and AutoCAD were completed at CADD Centre, India. Other skills are self-taught. Certificates are attached to the portfolio; scan the QR code to access them.